

Activity:

1. Data Cleaning

- Remove duplicate employee records.
- Handle missing or null salary values (e.g., replace with 0 or flag them).

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists 'Pipelines' (1), 'Data flows' (1), and 'Power Query' (0). The 'dataflow1' pipeline is selected. The main canvas displays a data flow diagram with the following steps: 'source1' (Columns: 4 total), 'dropdupli' (Aggregating data by 'EmployeeID' producing columns 'Name, Department, Salary'), 'derivedColumn1' (Creating/updating the columns 'EmployeeID, Name, Department, Salary, CleanSalary, CleanDept'), 'derivedColumn2' (Creating/updating the columns 'EmployeeID, Name, Department, Salary, CleanSalary, CleanDept'), and 'filter1' (Filtering rows using expressions on 'CleanSalary'). The 'Source settings' tab is active for the 'source1' step. It shows the 'Output stream name' as 'source1', 'Description' as 'Import data from data', 'Source type' as 'Dataset', and 'Dataset' as 'data'. The 'Options' section includes 'Allow schema drift' (checked), 'Infer drifted column types' (unchecked), and 'Validate schema' (unchecked). The 'Skip line count' is set to 0, and 'Sampling' is set to 'Disable'.

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Factory Resources

- Pipelines: 1
 - transform
- Change Data Capture (preview): 0
- Datasets: 2
 - AzureSqlTable
 - data
- Data flows: 1
 - dataflow1
- Power Query: 0

transform • data • dataflow1 • AzureSqlTable

Validate Data flow debug Debug Settings

source1 Import data from data

dropdupli Columns: 4 total

derivedColumn1 Creating/updating the columns EmployeeID, Name, Department, Salary, CleanSalary, CleanDept

derivedColumn2 Creating/updating the columns EmployeeID, Name, Department, Salary, CleanSalary, CleanDept

filter1 Filtering rows using expressions on 'CleanSalary'

filter1 Filtering rows using expressions on 'CleanSalary'

Aggregate settings Optimize Inspect Data preview

Aggregate settings

Description: Aggregating data by 'EmployeeID' producing columns 'Name, Department, Salary'

Incoming stream: source1

Group by: EmployeeID

Grouped by: EmployeeID

Column	Expression
Name	first(Name)
Department	first(Department)
Salary	first(Salary)

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 - dataflow1
- Power Query: 0

transform • data • dataflow1 • AzureSqlTable

Validate Data flow debug Debug Settings

source1 Import data from data

dropdupli Aggregating data by 'EmployeeID' producing columns 'Name, Department, Salary'

derivedColumn1 Columns: 6 total

derivedColumn2 Creating/updating the columns EmployeeID, Name, Department, Salary, CleanSalary, CleanDept

filter1 Filtering rows using expressions on 'CleanSalary'

filter1 Filtering rows using expressions on 'CleanSalary'

Derived column's settings Optimize Inspect Data preview

Derived column's settings

Output stream name: derivedColumn1

Description: Creating/updating the columns EmployeeID, Name, Department, Salary, CleanSalary, CleanDept

Incoming stream: dropdupli

Columns

Column	Expression
CleanSalary	if(isNull(Salary)) Salary == "", 0, toInteger(Salar...
CleanDept	upper(replace(trim(Department), ' ', '_'))

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1. Data Transformation

- Standardize department names (convert to uppercase, replace spaces with underscores).
- Calculate the **sum of salary per department**.
- Calculate the **average salary per department**.

2. Derived Columns

a) Add a new column **SalaryCategory** based on salary ranges (e.g., <3000 = **LowCount**, 3000–7000 = **MediumCount**, >7000 = **HighCount**).

3. Filtering

a) Exclude employees with salary = 0 after cleaning.

The screenshot shows the Microsoft Azure Data Factory interface. The left sidebar displays 'Factory Resources' including Pipelines, Datasets, Data flows, and Power Query. The main canvas shows a data flow named 'dataflow1' with the following steps: 'source1' (Import data from data), 'dropdupli' (Aggregating data by EmployeeID producing columns: Name, Department, Salary), 'derivedColumn1' (Creating/updating the columns EmployeeID, Name, Department, Salary, CleanSalary, CleanDept), and 'derivedColumn2' (Columns: 7 total). The 'Derived column's settings' pane for 'derivedColumn2' is open, showing the 'Columns' section with a table:

Column	Expression
SalaryCat	if(CleanSalary < 3000, 'LowCount', if(CleanSalary >= 3000 & CleanSalary <= 7000, 'MediumCount', if(CleanSalary > 7000, 'HighCount', '')))

The 'Output stream name' is 'derivedColumn2' and the 'Incoming stream' is 'derivedColumn1'. The 'Description' is 'Creating/updating the columns EmployeeID, Name, Department, Salary, CleanSalary, CleanDept, SalaryCat'.

The screenshot shows the Microsoft Azure Data Factory interface with the data flow 'dataflow1' updated. The steps are: 'dropdupli', 'derivedColumn1', 'derivedColumn2', 'filter1' (Columns: 7 total), and 'sink1' (Export data to AzureSqlTable). The 'Filter settings' pane for 'filter1' is open, showing the 'Filter on' section with a table:

Filter on
CleanSalary > 0

The 'Output stream name' is 'filter1' and the 'Incoming stream' is 'derivedColumn2'. The 'Description' is 'Filtering rows using expressions on columns: CleanSalary'.

Output

- a) Write the transformed data into Azure Postgres Database.
- b) Ensure the output contains:
 - i. Department
 - ii. Total Salary
 - iii. Average Salary
 - iv. Salary Category counts

The screenshot displays the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane shows a hierarchy: Pipelines (1), Change Data Capture (preview) (0), Datasets (2), Data flows (1), and Power Query (0). The 'dataflow1' data flow is selected. The main canvas shows a data flow diagram with the following components: 'dedupli' (deduplicating data by EmployeeID), 'derivedColumn1' (creating/updating columns: EmployeeID, Name, Department, Salary, CleanSalary, CleanDept), 'derivedColumn2' (creating/updating columns: EmployeeID, Name, Department, Salary, CleanSalary, CleanDept), 'filter1' (filtering rows using expressions on columns: CleanSalary), 'sink1' (exporting data to AzureSqlTable, 7 columns total), and 'aggregate1' (aggregating data by CleanDept, producing columns: TotalSalary, AvgSalary, LowCount, MediumCount). Below the diagram, the 'Sink' configuration pane is visible, showing the following settings:

- Output stream name: sink1
- Description: Export data to AzureSqlTable
- Incoming stream: filter1
- Sink type: Dataset
- Dataset: AzureSqlTable
- Options: ☒ Allow schema drift, ☐ Validate schema

An 'Activate Windows' watermark is visible in the bottom right corner of the interface.

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Factory Resources

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- AzureSqlTable
- data
- Data flows 1
- dataflow1
- Power Query 0

transform • data • dataflow1 • AzureSqlTable

Azure SQL Database
AzureSqlTable

Properties

General Related (1)

Name *
AzureSqlTable

Description

Annotations
+ New

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Cancel

Connection Schema Parameters

Linked service *
AzureSqlDatabase1 Test connection Edit + New Learn more

Table
schema name finaloutput
☒ Enter manually

Preview data

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Factory Resources

- Pipelines 1
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- dataflow1
- Power Query 0

transform • data • dataflow1 • AzureSqlTable

Validate Data flow debug Debug Settings

derivedColumn1
Creating/updating the columns
EmployeeID, Name,
Department, Salary,
CleanSalary, CleanDept

derivedColumn2
Creating/updating the columns
EmployeeID, Name,
Department, Salary,
CleanSalary, CleanDept

filter1
Filtering rows using
expressions on columns
CleanSalary

sink1
Export data to AzureSqlTable

filter1
Filtering rows using
expressions on columns
CleanSalary

aggregate1
Columns:
6 total

Aggregate settings Optimize Inspect Data preview

producing columns 'Totalsalary, AvgSalary,
LowCount, MediumCount, HighCount'

Incoming stream *
filter1

Group by Aggregates

Grouped by: CleanDept
+ Add Clone Delete Open expression builder

Column	Expression
Totalsalary	sum(CleanSalary)
AvgSalary	avg(CleanSalary)
LowCount	countif(SalaryCat == 'LowCount')
MediumCount	countif(SalaryCat == 'MediumCount')
HighCount	countif(SalaryCat == 'HighCount')

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Factory Resources

- Pipelines 1
- transform
- Change Data Capture (preview) 0
- Datasets 2
 - AzureSqlTable
 - data
- Data flows 1
 - dataflow1
- Power Query 0

transform • data • dataflow1 • AzureSqlTable

Validate Data flow debug Debug Settings

derivedColumn1 Creating/updating the columns: EmployeeID, Name, Department, Salary, CleanSalary, CleanDept

derivedColumn2 Creating/updating the columns: EmployeeID, Name, Department, Salary, CleanSalary, CleanDept

filter1 Filtering rows using expressions on columns 'CleanSalary'

sink1 Export data to AzureSqlTable

filter1 Filtering rows using expressions on columns 'CleanSalary'

aggregate1 Columns: 6 total

Aggregate settings Optimize Inspect Data preview

Number of rows: INSERT 6 UPDATE 0 DELETE 0 UPSERT 0 LOOKUP 0 ERROR 0 TOTAL 6

Refresh Typecast Modify Map drifted Statistics Remove Export to CSV

	CleanDept	Totalsalary	AvgSalary	LowCount	MediumCount	HighCount
+	OPERATIONS	889200	7288.524590163935	14	40	68
+	HR	645300	7170.0	10	32	48
+	SALES	790900	6818.103448275862	13	50	53
+	FINANCE	766100	7296.190476190476	12	38	55
+	MARKETING	774300	6975.675675675676	8	46	57
+	IT	731100	7311.0	8	43	49

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free-sql-formls (dbserverlms/free-sql-formls) | Query editor (preview) ☆

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Overview Activity log Tags Diagnose and solve problems Query editor (preview) Mirror database in Fabric (preview) Resource visualizer Settings Compute + storage Connection strings Maintenance Properties Locks Data management Replicas Sync to other databases Integrations Azure Synapse Link Stream analytics (preview) Add Azure AI Search Power Platform Power BI Power Apps Power Automate Security

Explorer

- free-sql-formls
 - dbo
 - Tables
 - finaloutput
 - Views
 - Stored Procedures
 - Functions
 - Queries
 - SQL query 1

SQL query 1

Run Save as view

```
1 select * from finaloutput;
```

Messages Results {}

	EmployeeID	Employee Name	Employee Department	Employee Salary	CleanSalary	CleanDept	SalaryCat
1	1884	Employee_1884	IT	11400	11400	IT	HighCount
2	1084	Employee_1084	sales	10800	10800	SALES	HighCount
3	1460	Employee_1460	Sales	3100	3100	SALES	MediumCount
4	1483	Employee_1483	hr	2200	2200	HR	LowCount
5	1507	Employee_1507	IT	10500	10500	IT	HighCount
6	1721	Employee_1721	Finance	2500	2500	FINANCE	LowCount
7	1088	Employee_1088	marketing	9200	9200	MARKETING	HighCount
8	1238	Employee_1238	Sales	3700	3700	SALES	MediumCount
9	1342	Employee_1342	Sales	7800	7800	SALES	HighCount
10	1591	Employee_1591	marketing	5400	5400	MARKETING	MediumCount
11	1645	Employee_1645	marketing	5100	5100	MARKETING	MediumCount
12	1829	Employee_1829	sales	5300	5300	SALES	MediumCount
13	1139	Employee_1139	finance	9700	9700	FINANCE	HighCount
14	1322	Employee_1322	operations	5500	5500	OPERATIONS	MediumCount
15	1352	Employee_1352	marketing	9300	9300	MARKETING	HighCount
16	1618	Employee_1618	Operations	8200	8200	OPERATIONS	HighCount
17	1650	Employee_1650	finance	4800	4800	FINANCE	MediumCount
18	1903	Employee_1903	operations	9200	9200	OPERATIONS	HighCount
19	1223	Employee_1223	marketing	2300	2300	MARKETING	LowCount
20	1005	Employee_1005	finance	4100	4100	FINANCE	MediumCount
21	1016	Employee_1016	marketing	7300	7300	MARKETING	MediumCount

Succeeded (0 sec 155 ms)

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